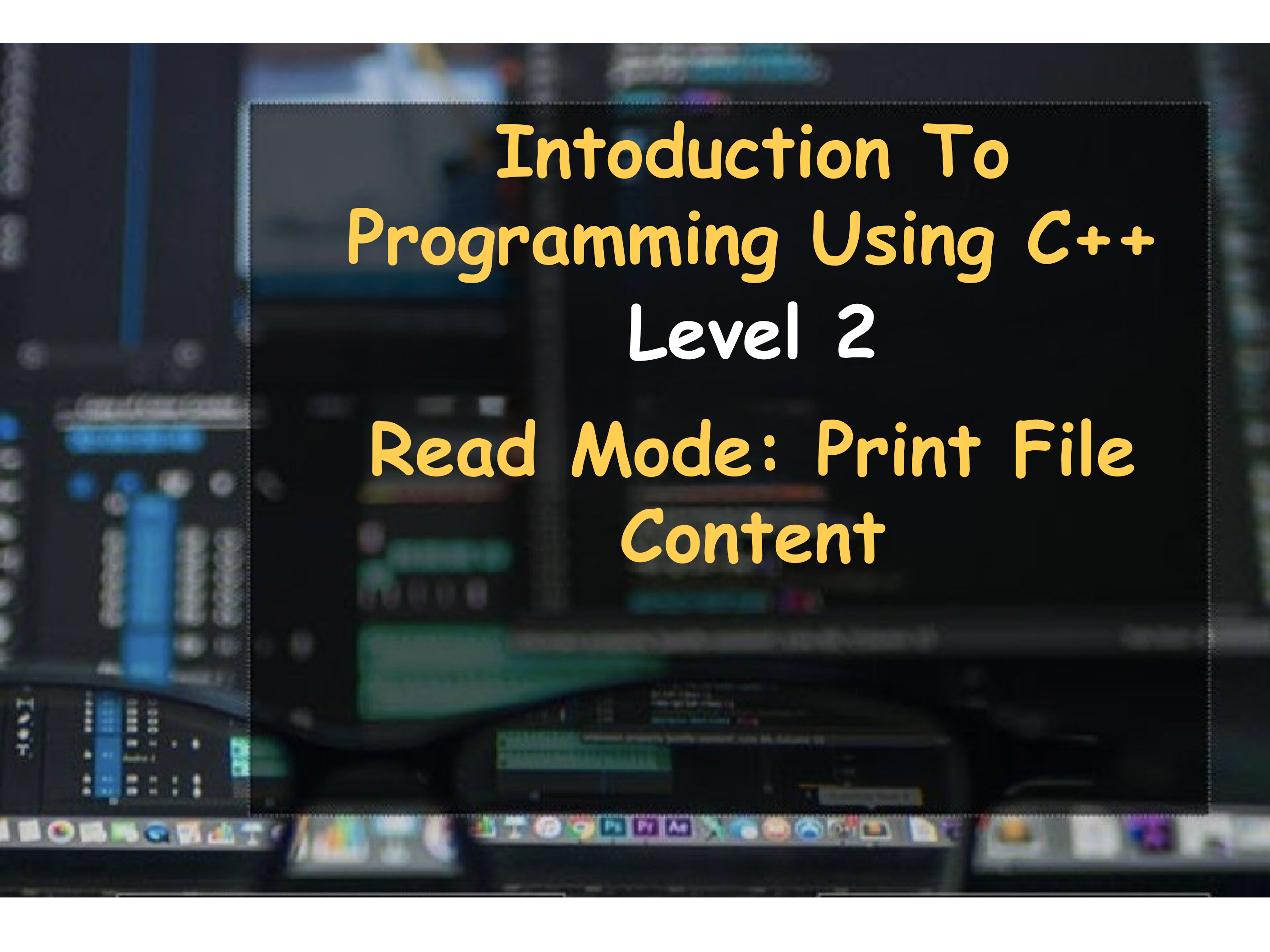
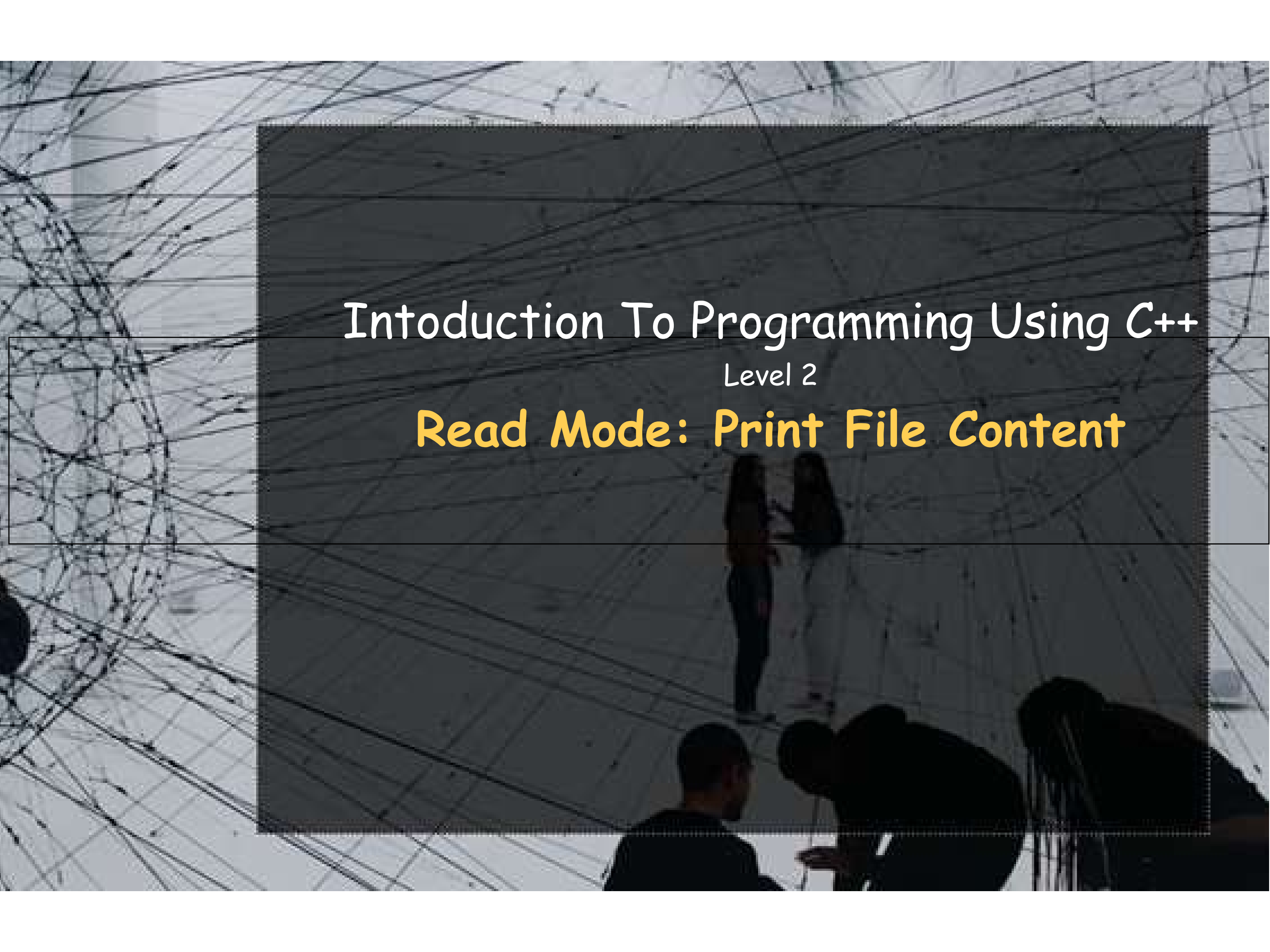




Introduction To Programming Using C++ Level 2

Read Mode: Print File Content



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Definition

Read Mode

 Read Mode in C++ is used to read the contents of an existing file and display the data on the console.

 It is used for reading data from the file, not for modifying, deleting, or appending data to it.

Explanation

`ios::in`

 `ios::in` means input and opens the file for reading.

```
File.open("FileName.txt", ios::in);
```

 We can also specify the complete path of the file:

```
File.open("C:/Documents/CPP/files/MyFile.txt", ios::in);
```

`is\_open()`

 We use `is\_open()` to check whether the file was opened successfully.

```
if (File.is_open())
```

`string Line`

 We create a string variable to store each line read from the file.

```
string Line;
```

`getline()`

 `getline(File, Line)` reads one complete line from the file and stores it in

`Line`.

 The `while` loop continues reading until there are no more lines to read.

```
while (getline(File, Line))
```

```
{
```

```
    cout << Line << endl;
```

```
}
```

`close()`

 After finishing the reading operation, we close the file.

```
File.close();
```

Procedures

1. Open the file using `ios::in`.
2. Check that the file was opened successfully using `is_open()`.
3. Create a string variable to store each line.
4. Use `getline()` inside a `while` loop to read the file line by line.
5. Print each line using `cout`.
6. Close the file using `close()`.

💡 Example

```
void PrintFileContent(string FileName)
{
    fstream MyFile;

    MyFile.open(FileName, ios::in);    // Read Mode

    if (MyFile.is_open())
    {
        string Line;

        while (getline(MyFile, Line))
        {
            cout << Line << endl;
        }
        MyFile.close();
    }
}

int main()
{
    PrintFileContent("MyFile.txt");
    return 0;
}
```

💡 Example

➡ Result

If `MyFile.txt` contains:

Hi, this is the first line

Hi, this is the second line

Hi, this is the third line

The program prints:

Hi, this is the first line

Hi, this is the second line

Hi, this is the third line

Note

- ◆ Reading from a file is similar to using ``cin`` , but instead of receiving input from the keyboard, we receive input from the file.
- ◆ ``getline()`` reads a complete line, including spaces.
- ◆ The ``while`` loop continues as long as ``getline()`` successfully reads another line.
- ◆ The file acts as a data source for the program.

⊘ Warning

- ◆ If you try to open a file that does not exist using ``ios::in``, the file is not created.
- ◆ Always check ``is_open()`` before trying to read from the file.
- ◆ Do not use ``ios::in`` when your intention is to modify or append data to the file.



Comparison

``ios::in``

- ✓ Used for reading.
- ✓ Does not modify the file contents.
- ✓ Does not create a missing file.
- ✓ Commonly used with ``getline()``.

``ios::out``

- ✓ Used for writing.
- ✓ Can create the file if it does not exist.
- ✓ Opening an existing file for output may truncate its contents.
- ✓ Commonly used with ``<<``.

✦ Summary

- ✓ ``ios::in`` = Read Mode.
- ✓ Used to read data from an existing file.
- ✓ ``is_open()`` checks whether the file opened successfully.
- ✓ ``getline()`` reads the file line by line.
- ✓ ``while (getline(...))`` continues until the end of the file.
- ✓ ``cout`` displays the read data on the console.
- ✓ ``close()`` closes the file after reading.
- ✓ ``ios::in`` does not create a file if it does not exist.

Thank you 😊